



Lab Title: OSI Model Lab Exercise

Student Name: Nebeuwa Ifeanyichukwu Raphael

Student ID: 202520850LE

Course Name: Fundamentals of Networking

Instructor Name: Zakariyya Dahuru Musa

Date of Submission: 27/10/2025

Executive Summary

This lab focuses on understanding the OSI (Open Systems Interconnection) model from a cybersecurity perspective. It explores how cyber-attacks target different layers of the model and the corresponding defense mechanisms that can be applied. Through research and analysis, the lab demonstrates that security must be implemented across all layers from the physical layer that deals with hardware and cabling to the application layer that supports user interactions and services. The exercise reinforces the importance of a defense-in-depth strategy, where multiple layers of security work together to prevent, detect, and respond to cyber threats effectively.

Lab Objectives

1. To identify and understand the seven layers of the OSI model and their functions in network communication.
2. To analyze how various types of cyber-attacks exploit vulnerabilities at different OSI layers.
3. To research and describe real-world examples of attacks targeting specific OSI layers.
4. To explore and document practical security controls and mitigation techniques for each OSI layer.
5. To develop a Cybersecurity Quick Reference Guide that maps layers to protocols, technologies, attacks, and mitigation strategies.

6. To strengthen analytical and research skills relevant to cyber security defense and network protection.

Tools and Resources

- ICDFFA cyber security lab materials.
- Online references from OWASP and Cisco Networking Academy.

Methodology

The lab exercise was carried out through research, analysis, and documentation using a structured approach:

1. **Layer Selection and Attack Research:**
Three OSI layers were selected for detailed analysis. For each, a real or hypothetical cyber-attack was researched to understand its mechanism and impact. Examples include SQL Injection (Layer 7), ARP Poisoning (Layer 2), and ICMP Flood (Layer 3).
2. **Threat and Control Analysis:**
For every identified attack, the specific vulnerability exploited was explained, followed by appropriate security controls or mitigation measures, such as firewalls, encryption, or input validation.
3. **Cybersecurity Quick Reference Guide Development:**
Three structured tables were completed to map each OSI layer's core function, security focus, associated protocols, and tools used for protection. Additionally, four layers were analyzed in terms of attack type and mitigation strategies to reinforce the practical connection between theory and security application.
4. **Compilation and Reporting:**
All findings were summarized and formatted into a professional PDF report following cyber security documentation standards for academic and practical use.

Part 1: Research & Analysis - The Attack of the Layers

Scenario: A company called "SecureCorp" was hit by a sophisticated cyber-attack. The attack involved multiple stages, each targeting a different layer of the OSI model. Your task is to research and describe a real or hypothetical cyber-attack that exploits vulnerabilities at **three different OSI layers**.

1. Layer 7 - Application

- **Attack Name:** SQL Injection, Cross-Site Scripting (XSS), Phishing, DDoS (HTTP Flood)
- **How the Attack Works:** The attacker inserts malicious SQL code into an application's input fields (like a login form or search bar). When the server processes it, the code manipulates the database - allowing the attacker to steal, modify, or delete data.
- **Security Control/Mitigation:** Use input validation and prepared SQL statements to ensure user input cannot alter database queries.

2. Layer 6 - Presentation

- **Attack Name:** Malware in File Attachments, SSL/TLS Attacks
- **How the Attack Works:** Hackers exploit data encoding or weak encryption to intercept or corrupt data.
- **Security Control/Mitigation:** Enforce HSTS (HTTP Strict Transport Security) and disable weak encryption protocols (like SSLv2/v3) to ensure only secure HTTPS connections are allowed.

3. Layer 5 - Session

- **Attack Name:** Session Hijacking, Session Fixation
- **How the Attack Works:** Attackers steal or reuse your login session to act as you.
- **Security Control/Mitigation:** Use secure tokens, HTTPS, and automatic session expiration.

Part 2: OSI Layer Security Reference Tables

Your goal is to create a comprehensive "Cybersecurity Quick Reference Guide" for the OSI model. Complete the following three tables with the required details.

OSI Layer Number	Layer Name	Key Function	Primary Cybersecurity Focus
7	Application	Provides network services directly to user applications (e.g., HTTP, FTP, DNS).	Protecting the applications and data from malicious input and unauthorized access.
6	Presentation	Translates, encrypt, and compresses data	Protecting the integrity and confidentiality of the data format itself (e.g., ensuring encryption is strong, data isn't tampered with during translation).
5	Session	Establish, manages and terminates communication session between application	Protecting the session dialog, preventing unauthorized parties from hijacking or taking over an established session.
4	Transport	Provides reliable, end-to-end data transport and flow between hosts (using TCP/UDP)	Protecting the "conversation" between two hosts, ensuring it is not interrupted, hijacked, or flooded with traffic (Denial-of-Service).
3	Network	Manages logical addressing (IP) and routing	Protecting the routing path and logical addressing, preventing packets from being misdirected, intercepted, or spoofed.
2	Data Link	Handles node-to-node data transfer on the same physical link, using MAC address	Protecting the local network segment, ensuring frames are delivered only to the intended physical device and not intercepted on the local LAN
1	Physical	Transport the raw bit stream (electrical, light, radio signals) across the network medium	Protecting the physical infrastructure itself from theft, tampering, or destruction that would interrupt the binary signal.

OSI Layer Number	Layer Name	Example Protocols & Technologies	A Cybersecurity Technology/Tool that operates at this layer
7	Application	HTTP, HTTPS, DNS, FTP, SMTP	Web Application Firewall (WAF), DLP Systems
6	Presentation	SSL/TLS, JPEG, MPEG, ASCII, PDF	SSL/TLS Inspection (in Firewalls/IPS), Encryption/Decryption Software
5	Session	NetBIOS, RPC, PAP, SCP, SIP	Session Monitoring Tools, Honeypots (to trap session-based attacks)
4	Transport	TCP, UDP	Stateful Firewall, Intrusion Prevention System (IPS), Load Balancer (for DoS protection)
3	Network	IP (IPv4, IPv6), ICMP, IPsec, Routers, BGP, OSPF	Network Firewall, Router ACLs, IPsec VPN, Ping/ICMP Blocking
2	Data Link	PPP, ARP, VLANs, Switches, MAC Addresses, Frame Relay	Switch Port Security, ARP Inspection, MAC Address Filtering, VLAN Access Control Lists (VACLs)
1	Physical	1 Cat5/6/7 Cables, Fiber Optics, Hubs, Repeaters, Wireless Radio Frequencies, NICs	Physical Locks, Cabinet Locks, Biometric Access Controls, Cable Shielding, Network Tap (for monitoring)

Table 2: Protocol & Technology Mapping

Table 3: Attack & Mitigation Mapping

OSI Layer Number	Layer Name	One Specific Cyber Attack (from Part 1 or another)	One Appropriate Mitigation Strategy
2	Data Link	ARP Poisoning	Use static ARP entries where possible, dynamic ARP inspection, DHCP snooping.
3	Network	IP Spoofing	Ingress/egress filtering (BCP 38), anti-spoofing controls.
6	Presentation	SSL/TLS Attacks	Enforce HSTS and disable weak SSL/TLS protocols.
7	Application	Phishing	Input validation and prepared statements to prevent injection.

Reference

- **International Cybersecurity and Digital Forensic Academy (ICDFA).** *Cybersecurity Lab Materials and Exercises*. Retrieved from [ICDFA Learning Portal or Course Materials]
- **OWASP Foundation.** (2025). *OWASP Top Ten Web Application Security Risks*. Retrieved from <https://owasp.org>
- **Cisco Networking Academy.** (2025). *Introduction to Networking and Cybersecurity*. Retrieved from <https://www.netacad.com>